

UCT101: DISCRETE MATHEMATICS

L	T	P	Cr
3	1	0	3.5

Course Objectives: Detailed study of various discrete and algebraic structures, basic logic, basics of counting and proof techniques.

Boolean algebra: Introduction of Boolean algebra, truth table, basic logic gate, basic postulates of Boolean algebra, principle of duality, canonical form, Karnaugh map. Abstract algebra: Set, relation, group, ring, field.

Combinatorics: Basic counting, balls and bins problems, generating functions, recurrence relations. Proof techniques, principle of mathematical induction, pigeonhole principle.

Graph Theory: Graphs and digraphs, complement, isomorphism, connectedness and reachability, adjacency matrix, Eulerian paths and circuits in graphs and digraphs, Hamiltonian paths and circuits in graphs and tournaments, trees; Planar graphs, Euler's formula, dual of a planar graph, independence number and clique number, chromatic number, statement of Four-color theorem.

Logic: Propositional calculus - propositions and connectives, syntax; Semantics - truth assignments and truth tables, validity and satisfiability, tautology; Adequate set of connectives; Equivalence and normal forms; Compactness and resolution; Formal reducibility - natural deduction system and axiom system; Soundness and completeness.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

The students will be able to reflect on:

1. Perform operations on various discrete structures such as set, function and relation.
2. Evaluate Boolean functions and simplify expression using the properties of Boolean algebra.
3. Illustrate the basic properties and algorithms of graphs and apply them in modeling and solving real-world problems.
4. Comprehend formal logical arguments and translate statements from a natural language into its symbolic structures in logic.
5. Identify and prove various properties of rings, fields and group.

Text Books:

1. Rosen H. K., Discrete Mathematics and its Applications, McGraw Hill (2011) 7th ed.
2. Tremblay P. J. and Manohar, R., Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill (2008).
3. Topics in Algebra, I. N. Herstein, John Wiley and Sons.
4. Digital Logic & Computer Design, M. Morris Mano, Pearson.
5. Elements of Discrete Mathematics, (Second Edition) C. L. Liu McGraw Hill, New Delhi.
6. Graph Theory with Applications, J. A. Bondy and U. S. R. Murty, Macmillan Press, London.
7. Mathematical Logic for Computer Science, L. Zhongwan, World Scientific, Singapore.

Reference Books:

1. Gallian A. J., Contemporary Abstract Algebra, Cengage Learning (2017) 9th ed.
2. Lipschutz S., Lipson M., Discrete Mathematics, McGraw-Hill (2007) 3rd ed.
3. Introduction to linear algebra. Gilbert Strang.
4. Introductory Combinatorics, R. A. Brualdi, North-Holland, New York.
5. Graph Theory with Applications to Engineering and Computer Science, N. Deo, Prentice Hall, Englewood Cliffs.
6. Introduction to Mathematical Logic, (Second Edition), E. Mendelsohn, Van-Nostrand, London.